

Analysis of LOS

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Analysis of Traffic LOS by Facility Type

Steps

- 1. Find flow (v_p) of the traffic stream
- 2. Find speed (S) of the traffic Stream
- 3. Find density of traffic stream (D) using $v_p = D \times s$
- 4. Check appropriate curve for LOS

1. Find flow (v_p) of the traffic stream

- $$v_p = \frac{V}{PHF \times f_{HV} \times N}$$
 - v_p = flow rate per lane under equivalent ideal condition (pc/hr/ln)
 - V = demand volume (veh/hr)
 - PHF = peak-hour factor
 - f_{HV} = adj. factor for presence of heavy vehicles (HV)
 - N = number of lanes in one direction

1.2 Adj. factor for presence of heavy vehicles (HV)

- $f_{HV} = \frac{1}{1 + P_{HV}(E_{HV} - 1)}$
 - f_{HV} = adj. factor for presence of heavy vehicles (HV)
 - P_{HV} = proportion of HV in traffic stream
 - E_{HV} = heavy vehicle equivalent
 - $E_{HV} \sim f(\textit{Type of Terrain, Grade})$

1.2.1 Passenger Car Equivalents (E_{HV})

- $V_{pc} = V \times P_{HV} \times E_{HV}$
 - V_{pc} = volume in traffic stream in (pc/hr)
 - V = volume in traffic stream (veh/hr)
 - P_{HV} = proportion of HV in traffic stream
 - E_{HV} = heavy vehicle equivalent

1.2.2 E_{HV} for Terrain and Grades

- General Terrain Segments: Table 28.17
 - Condition:
 - NO one grade ≥ 3 longer than 0.25 *mile*
 - and NO one grad < 3 longer than 0.50 *mile*
 - Level terrain
 - Rolling terrain
- Specific Grades:
 - Typical Mix of HV: Table 28.18
 - Atypical Mix of HV: Table 28.19
 - Composite Grade:
 - To find equivalent grade for a series of upgrades and/or downgrades of varying steepness.
 - Average Grade, $G_{AV} = (\frac{\textit{total rise}}{\textit{total length}}) \times 100$
 - Use G_{AV} in Tables 28.18 and 28.19

2. Find speed (S) of the traffic Stream

$v_p \leq BP$	$v_p > BP$
$S = FFS_{adj}$	$S = FFS_{adj} - [\frac{(FFS_{adj} - \frac{c_{adj}}{4})(v_p - BP)^a}{(c_{adj} - BP)^a}]$

where,

- FFS_{adj} = adjustment free-flow speed (mph)
- BP = Breakpoint flow rate (pc/h/ln)
- c_{adj} = adjusted capacity (pc/h/ln)
- a = calibration constant

2.1 Adjusted Free-Flow Speed FFS_{adj}

Freeways	Multilane Highway
$FFS_{adj} = FFS \times SAF$	$FFS_{adj} = FFS$

where

- FFS_{adj} = Adjusted Free-Flow Speed
 - FFS = Free-Flow Speed
 - SAF = Speed Adjustment factor

2.1.1 Free-Flow Speed FFS

Freeways	Multilane Highway
$FFS = 75.4 - f_{LW} - f_{LC} - 3.22 \times TRD^{0.84}$	$FFS = BFFS - f_{LW} - f_{LC} - f_M - f_A$

Where

- FFS = free-flow spee of the freeway (mph)
- f_{LW} = adjustment for lane width (mph)

- f_{LC} = adjustment for right-side lateral clearance (mph)
- TRD = total ramp density (ramps / mile)
- f_M = adjustment for type of median (mph)
- f_A = adjustment for access points (mpg)

Factor	Freeways Table	Multilane Hwy Table
f_{LW}	Table 28.3	Table 28.3
f_{LC}	Table 28.4	Table 28.5
f_M	-	Table 28.6
f_A	-	Table 28.7

2.1.2 Speed Adjustment Factor SAF

$SAF = SAF_w \times SAF_{DP} \times SAF_{wz}$

where

- SAF = Speed adjustment factor
 - SAF_W = SAF for weather
 - SAF_{DP} = SAF for non-std driver poulation
 - SAF_{WZ} = SAF for work zone
 - $SAF_{WZ} = \frac{FFS_{WZ}}{FFS}$

Factor	Table
SAF_W	Table 28.11
SAF_{DP}	Table 28.13
SAF_{WZ}	-

- $SAF_{WZ} = \frac{FFS_{WZ}}{FFS}$
 - $FFS_{WZ} = 9.95 + (33.49 \times f_{SR}) + (0.53 \times SL_{WZ}) - (5.60 \times LCSI) - (3.84 \times f_{BR}) - (1.71 \times f_{DN}) - (8.7 \times TRD)$
 - f_{SR} - speed limit ratio = $\frac{SL}{SL_{WZ}}$
 - SL = base speed limit of the freeway (mph)
 - SL_{WZ} = speed limit at work zone (mph)
 - $LCSI$ = Lane closure severity index
 - Table 28.14
 - f_{BR} = adj for barrier type;
 - 0: Concrete or other hard barrier
 - 1: cones or soft barrier
 - f_{DN} = adj for day or night
 - 0: day
 - 1: night
 - TRD = total ramp density (ramps / mi)

2.2 Breakpoint value, BP

- BP Values for facility type

Freeways	Multilane Highway
$BP = [1000 + 40(75 - FFS_{adj})] \times CAF^2$	$BP = 1,400 pc/h/ln$

Where

- BP = Breakpoint flow
 - FFS_{adj} = adjustment free-flow speed (mph)
 - CAF = Capcity adjustment factor

2.1.1 Adjustment free-flow speed, FFS_{adj}

- Same as section 1.1 ### 2.1.2 Capacity Adjustment Factor CAF

$CAF = CAF_w \times CAF_I \times CAF_{DP} \times CAF_{wz}$

where

- CAF = Speed adjustment factor
 - CAF_W = CAF for weather
 - CAF_I = CAF for incidents
 - CAF_{DP} = CAF for non-std driver poulation
 - CAF_{wz} = CAF for work zone
 - $CAF_{wz} = \frac{c_{wz}}{c}$

Factor	Table
CAF_W	Table 28.10
CAF_I	Table 28.12
CAF_{DP}	Table 28.13
CAF_{wz}	-

- $CAF_{wz} = \frac{c_{wz}}{c}$
- Capacity of work zone, $c_{WZ} = QDR_{WZ} \frac{100}{(100 - \alpha_{WZ})}$
 - QDR_{WZ} = queue discharge rate from work zone
 - $QDR_{WZ} = 2,093 - (154 \times LCSI) - (194 \times f_{BR}) - (179 \times f_{AT}) + (9 \times f_{LAT}) - (59 \times f_{DN})$
 - $LCSI$ = Lane closure severity index
 - Table 28.14
 - f_{BR} = adj for barrier type;
 - 0: Concrete or other hard barrier
 - 1: cones or soft barrier
 - f_{AT} = adj for area type
 - 0: Urban
 - 1: Rural
 - f_{LAT} = adj for lateral clearance to barrier

- (0 - 12)ft
- f_{DN} = adj for day or night
 - 0: day
 - 1: night
- α_{WZ} = % reduction from capacity to QDR where

2.3 Adjusted Capacity, c_{adj}

Freeways	Multilane Highway
$c_{adj} = c \times CAF$	$c_{adj} = c$
$c = 2200 + 10(FFS - 50)$	$c = 1900 + 20(FFS - 45)$
$c \leq 2,400\text{ pc/h/ln}$	$c \leq 2,300\text{ pc/h/ln}$
and	and
$55 \leq FFS \leq 75$	$45 \leq FFS \leq 70$

2.4 Calibration Constant, a

Freeways	Multilane Highway
$a = 2.00$	$a = 1.31$

3. Find density of traffic stream (D)

$$v_p = D \times s$$

4. Check appropriate curve for LOS

- For Freeways: Figure 28.6
- For Multilane Highways: Figure 28.7